- *Suppose $F(x)$ is the distribution function of a continuous random variable $X$. Then the probability density function $f(x)$ is given by*

$$f(x) = \frac{dF(x)}{dx} = F'(x)$$ *, whenever derivative exists.*

- *Suppose $X$ is a random variable with probability mass or density function $f(x)$* ***The expected value or mean or mathematical expectation of $X$,*** denoted by $E(x)$ or $\mu$ *is*

$$E(X) = \begin{cases} \sum_x xf(x) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} xf(x)\,dx & \text{if } X \text{ is continuous} \end{cases}$$

- *The* ***variance*** *of the random variable $X$* denoted by $V(X)$ or $\sigma^2$ (or $\sigma_x^2$) is

$$V(X) = E(X - E(X))^2 = E(X - \mu)^2$$

## Properties of Mathematical expectation and variance

(i) $E(aX + b) = aE(X) + b$, where $a$ and $b$ are constants

| | |
|---|---|
| Corollary 1: $E(aX) = aE(X)$ | ( when $b = 0$) |
| Corollary 2: $E(b) = b$ | (when $a = 0$) |

(ii) $Var(X) = E(X^2) - (E(X))^2$

(iii) $Var(aX + b) = a^2 Var(X)$ where $a$ and $b$ are constants

| | |
|---|---|
| Corollary 3: $V(aX) = a^2V(X)$ | (when $b = 0$) |
| Corollary 4: $V(b) = 0$ | (when $a = 0$) |

- Let $X$ be a random variable associated with a Bernoulli trial by defining it as $X$ (success) = 1 and $X$ (failure) = 0, such that

$$f(x) = \begin{cases} p & x = 1 \\ q = 1 - p & x = 0 \end{cases} \quad \text{where } 0 < p < 1$$

- $X$ is called a Bernoulli random variable and $f(x)$ is called the Bernoulli distribution.

- *If $X$* is a Bernoulli's random variable which follows Bernoulli distribution with parameter $p$, the mean $\mu$ and variance $\sigma^2$ are

$$\mu = p \quad \text{and} \quad \sigma^2 = pq$$



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